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Questions

Q1.

Heptane, C_7H_{16}, is one of the compounds present in crude oil.

(a) When heptane is reformed, the products include 2,2,3-trimethylbutane and cycloheptane.

(i) Give a reason why petrol should **not** contain a high proportion of heptane.

(1)

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(ii) Draw the **skeletal** formula of 2,2,3-trimethylbutane.

(1)

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(iii) Write the equation for reforming heptane into cycloheptane.
Use **molecular** formulae.
State symbols are not required.

(1)

(iv) When petrol is burned in a car engine, oxides of nitrogen are formed.
Explain how these compounds result in damage to trees.

(2)

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(b) Heptane reacts with chlorine in sunlight.

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(i) Chlorine radicals are formed in the first step in the mechanism.

$$Cl_2 \rightarrow 2Cl\bullet$$

Name this step in the mechanism.

(1)

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(ii) Give the **two** propagation steps for the formation of chloroheptane.
Use molecular formulae. Curly arrows are **not** required.

(2)

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(iii) Give the termination step which forms a hydrocarbon.

(1)

(iv) Explain how some dichloroheptane, $C_7H_{14}Cl_2$, also forms during this reaction.
You may include equation(s) in your answer.

(2)

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(Total for question = 11 marks)

Q2.

Members of the homologous series of alkanes have the same

- ☒ **A** boiling temperature
- ☒ **B** density
- ☒ **C** empirical formula
- ☒ **D** general formula

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(Total for question = 1 mark)

Q3.

This question is about cycloalkanes.

(a) When alkanes from crude oil are reformed, the products include cycloalkanes.

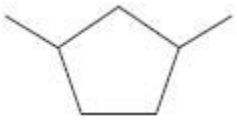
Write the equation for reforming hexane into cyclohexane using **skeletal** formulae for the organic compounds.

(2)

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(b) The skeletal formula of cycloalkane **D** is shown.



(i) Give the name of **D**.

(1)

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(ii) Give the molecular formula of **D**.

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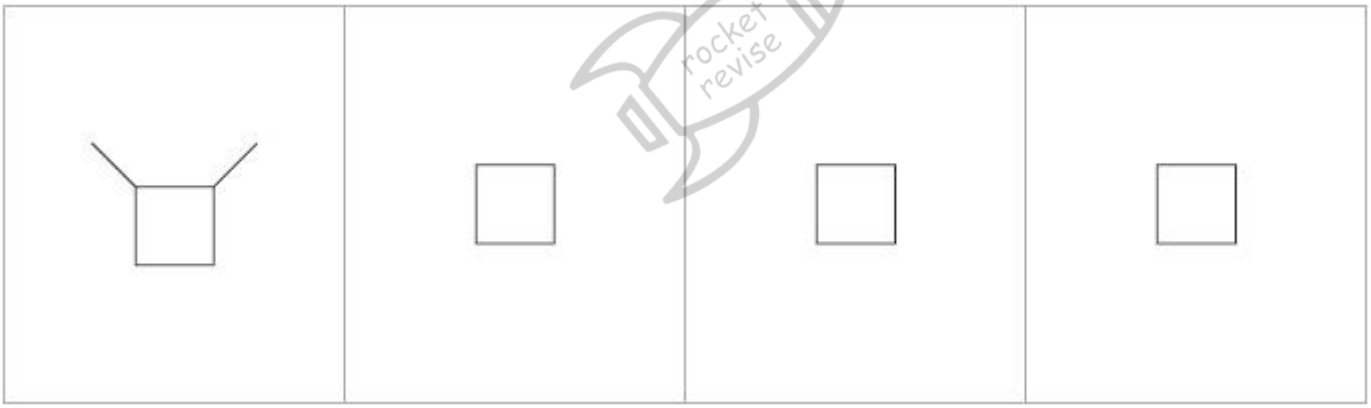
(1)

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(c) There are four structural isomers of C_6H_{12} with a ring of four carbon atoms.

One of these isomers is shown, in the first box.
Complete the **skeletal** formulae of the other three isomers.

(2)



(d) A cycloalkane, **E**, has a molar mass of 126 g mol^{-1}.

Deduce the molecular formula of **E**.

(1)

.....

(e) A sample of gaseous cyclopentane with a volume of 25 cm^3 was mixed with 250 cm^3 of oxygen (an excess) and the mixture was ignited.

Only gaseous products were formed.

$$2C_5H_{10}(g) + 15O_2(g) \rightarrow 10CO_2(g) + 10H_2O(g)$$

Calculate the volume of each gas remaining after the reaction.
All the gas volumes were measured at the same temperature and pressure.

(3)

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(f) Cyclobutane, C_4H_8, reacts with chlorine in sunlight.

(i) Name the mechanism and type of reaction that is occurring.

(2)

.....

(ii) Complete the equation for the initiation step of this reaction mechanism.
Include appropriate curly arrows.

(2)



(iii) Write the equations for the **two** propagation steps to form chlorocyclobutane.
Use C_4H_8 as the formula for cyclobutane.
Curly arrows and state symbols are not required.

(2)

(iv) A small amount of a hydrocarbon forms in this reaction.
Deduce the **skeletal** formula of this hydrocarbon. Justify your answer.

(2)

Skeletal formula of product

Justification

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.....

(Total for question = 18 marks)

Q4.

What is the total number of structural isomers with the molecular formula C_6H_{14}?

☒ **A** 4

☒ **B** 5

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☒ **C** 6

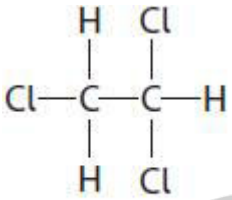
☒ **D** 7

(Total for question = 1 mark)

Q5.

In the upper atmosphere, ozone can react with halogens and with halogen-containing compounds, such as molecule **X**.

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molecule **X**

(a) Name molecule **X**.

(1)

.....

(b) The C—Cl bond undergoes homolytic fission in the upper atmosphere.

(i) State the essential condition necessary for this process.

(1)

.....

(ii) Write an equation for this fission in molecule **X**.
Include curly half-arrows.

(2)

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(iii) Write an equation for the reaction of a chlorine free radical with a molecule of **X**.
Curly half-arrows are not required.

(2)

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(iv) Write an equation, using displayed formulae, for a free radical formed in this sequence to give a molecule with the formula $C_4H_4Cl_6$.

(2)

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(v) State the type of reaction occurring in (b)(iv) and the name of the product with the formula $C_4H_4Cl_6$.

(2)

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(Total for question = 10 marks)

Q6.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the empirical formula of butane?

- ☐ **A** C_4H_{10}
- ☐ **B** C_2H_5
- ☐ **C** $CH_{2.5}$
- ☐ **D** C_nH_{2n+2}

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(Total for question = 1 mark)

Q7.

Heptane, C_7H_{16}, is an alkane found in crude oil.

(a) Heptane can undergo incomplete combustion.

(i) Give a reason why incomplete combustion sometimes occurs.

(1)

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(ii) Write the equation for the incomplete combustion of heptane, forming carbon monoxide and water as the
only products.
State symbols are not required.

(1)

(b) Heptane is reformed into branched-chain and cyclic hydrocarbons that are used in petrol.

(i) Draw the **skeletal** formulae of a branched-chain alkane and a cycloalkane, each containing **seven** carbon atoms.

(2)

Branched-chain alkane
Cycloalkane

(ii) Write the equation for the reforming of heptane into a cycloalkane.
Use molecular formulae.

(1)

(iii) Give a reason for adding cycloalkanes to petrol.

(1)

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(c) Heptane, C_7H_{16} , reacts with chlorine in the presence of ultraviolet radiation.

(i) State the type and mechanism of this reaction.

(2)

.....

(ii) Give the mechanism for the reaction to produce $C_7H_{15}Cl$, $C_{14}H_{30}$ and HCl as the **only** products.
Include the name of each of the steps in your mechanism.
Curly half-arrows are **not** required.

(7)

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(Total for question = 15 marks)

Q8.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

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Cyclopentane undergoes free radical substitution with bromine.

(a) Which of these is an overall equation for this reaction?

(1)

- ☐ **A** $C_5H_8 + Br_2 \rightarrow C_5H_8Br_2$
- ☐ **B** $C_5H_{10} + Br_2 \rightarrow C_5H_{10}Br_2$
- ☐ **C** $C_5H_{10} + Br_2 \rightarrow C_5H_8Br_2 + H_2$
- ☐ **D** $C_5H_{10} + Br_2 \rightarrow C_5H_9Br + HBr$

(b) Which statement is **not** correct about this reaction system?

(1)

- ☐ **A** only the initiation step involves homolytic bond fission
- ☐ **B** only some bromine is converted to free radicals in the initiation step

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☐ **C** propagation forms more product than termination
☐ **D** further substitution reactions are likely to occur

(c) Which free radical is **least** likely to form in a propagation step in this reaction system?

(1)

☐ **A** $C_5H_9^{\bullet}$

☐ **B** $Br^{\bullet}$

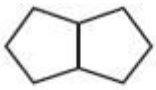
☐ **C** $C_5H_8Br^{\bullet}$

☐ **D** $H^{\bullet}$

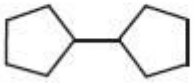
(d) Which alkane could be formed in a termination step in this reaction system?

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(1)

☐ **A** 

☐ **B** 

☐ **C** 

☐ **D** 

(Total for question = 4 marks)

Q9.

Methane reacts with excess chlorine in UV light.

(a) Which process occurs in the initiation step?

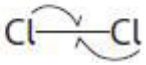
(1)

☐ **A** 

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☐ **B** 

☐ **C** 

☐ **D** 

(b) Which of these molecules could **not** be formed in a termination step?

(1)

☐ **A** C_2H_6
☐ **B** CH_3Cl

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☒ **C** CH_2Cl_2
☒ **D** HCl

(Total for question = 2 marks)

Q10.

Aluminium is an abundant metal with many uses.

(a) The first four ionisation energies of aluminium are shown.

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Ionisation number	Energy / kJ mol^{-1}
1	578
2	1820
3	2750
4	11 600

Explain how this information shows that aluminium is in Group 3 of the Periodic Table.

(2)

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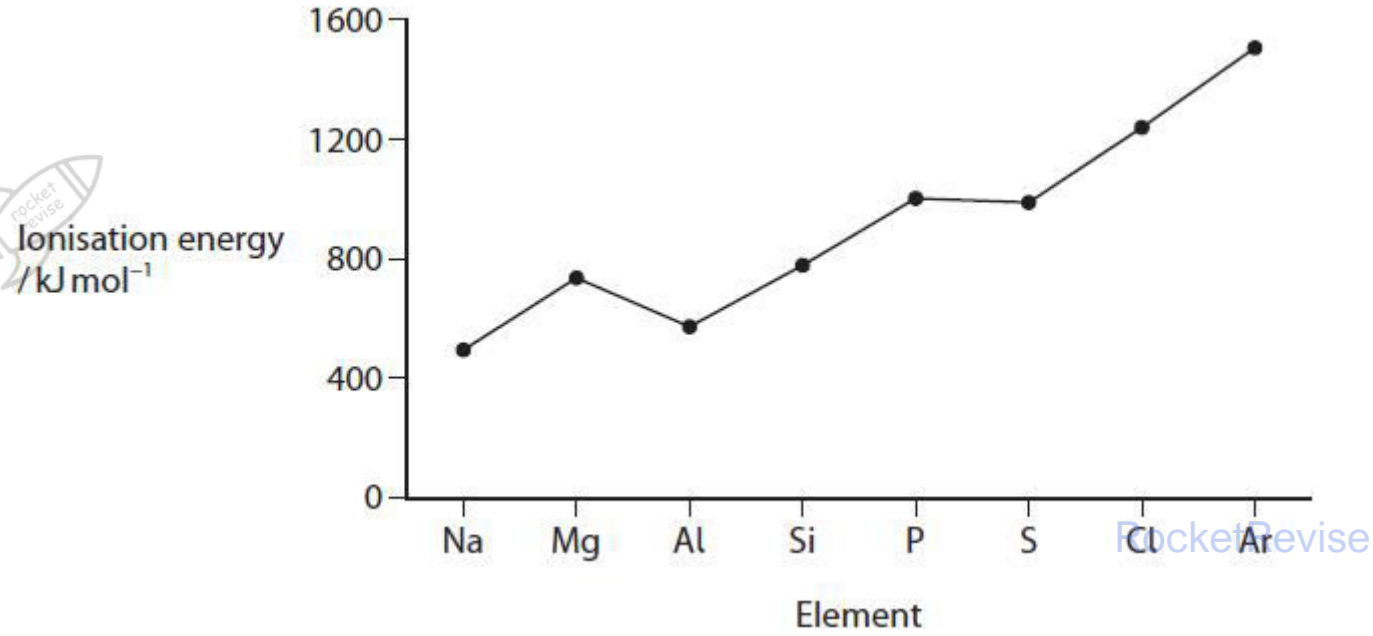
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(b) The graph shows the first ionisation energies for the elements in Period 3.

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(i) Explain the general increase in the first ionisation energy from sodium to argon.

(2)

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(ii) Explain why the first ionisation energy of aluminium is less than the first ionisation energy of magnesium.

(2)

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(c) (i) Describe the bonding in aluminium metal.

(2)

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(ii) Give two possible reasons why aluminium is used for overhead power cables.

(2)

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(d) New uses for waste aluminium cans are being investigated. One possible use is to make nanoparticle alloys to produce hydrogen for fuel.

(i) Aluminium nanoparticles react with water to produce aluminium oxide and hydrogen.
Complete the following equation. State symbols are not required.

(1)

..... Al + ... H_2O → ..... + ...

(ii) Give **two** possible reasons for producing hydrogen from aluminium rather than from fossil fuels.

(2)

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(Total for question = 13 marks)

Q11.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

When $C_{20}H_{42}$ is cracked, each molecule produces one molecule of ethene, one molecule of butane and two molecules of hydrocarbon **E**.

What is the molecular formula of **E**?

- ☐ **A** C_7H_{13}
- ☐ **B** C_7H_{14}
- ☐ **C** $C_{14}H_{26}$
- ☐ **D** $C_{14}H_{28}$

Use this space for any rough working. Anything you write in this space will gain no credit.

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(Total for question = 1 mark)

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Q12.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Petrol, bioethanol and hydrogen are fuels.

All three of these fuels

☐ **A** burn to produce greenhouse gases

☐ **B** are overall carbon neutral

☐ **C** are overall sustainable

☐ **D** biodegrade rapidly

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(Total for question = 1 mark)

Q13.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

How many structural isomers have the formula $C_3H_6Cl_2$?

☐ **A** 2
☐ **B** 3
☐ **C** 4
☐ **D** 5

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(Total for question = 1 mark)

Q14.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

All alkanes have the same

☐ **A** empirical formula

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- ☒ **B** general formula
- ☒ **C** molecular formula
- ☒ **D** structural formula

(Total for question = 1 mark)

Q15.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these atmospheric pollutants is **not** emitted during the combustion of alkane car fuels?

- ☒ **A** ammonia
- ☒ **B** nitrogen dioxide
- ☒ **C** sulfur dioxide
- ☒ **D** octane

(Total for question = 1 mark)

Q16.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A single molecule of decane, $C_{10}H_{22}$, is cracked.

Which of these mixtures could **not** be formed?

- ☒ **A** pentene and pentane
- ☒ **B** ethene, butene and butane
- ☒ **C** propene, propane and butene
- ☒ **D** hexene and propane

(Total for question = 1 mark)

Q17.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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When carrying out chemical experiments, the hazards and risks must be considered.

For a given chemical

- ☒ **A** the hazard is fixed but the risk varies
- ☒ **B** the hazard varies but the risk is fixed
- ☒ **C** both hazard and risk are fixed
- ☒ **D** both hazard and risk vary

(Total for question = 1 mark)

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Q18.

A 0.210 g sample of a volatile organic liquid **C** is injected into a gas syringe and heated in an oven.
At 100 kPa and 358 K, the syringe contains 72.5 cm^3 of gas.

(a) Calculate the molar mass of **C**.

[$pV = nRTR$ = 8.31 J mol^{-1} K^{-1}]

(4)

(b) The organic liquid **C** is a hydrocarbon.

Give a possible name or formula for **C**, using your answer in (a).

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(1)

(Total for question = 5 marks)

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Q19.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Ethane reacts with bromine in the presence of ultraviolet radiation.

(a) What is the equation for the reaction?

(1)

- ☒ **A** $C_2H_6 + Br_2 \rightarrow C_2H_4Br_2 + H_2$
- ☒ **B** $C_2H_6 + Br_2 \rightarrow C_2H_5Br + HBr$
- ☒ **C** $C_2H_6 + Br_2 \rightarrow 2CH_3Br$
- ☒ **D** $C_2H_6 + Br_2 \rightarrow CH_4 + CH_2Br_2$

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(b) The ultraviolet radiation is needed for

(1)

- ☒ **A** homolytic breaking of a Br—Br bond
- ☒ **B** heterolytic breaking of a Br—Br bond
- ☒ **C** homolytic breaking of a C—H bond
- ☒ **D** heterolytic breaking of a C—H bond

(Total for question = 2 marks)

Q20.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

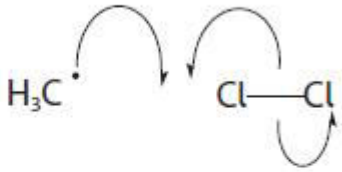
Which of these occurs in a **propagation** step in the reaction of methane with chlorine?

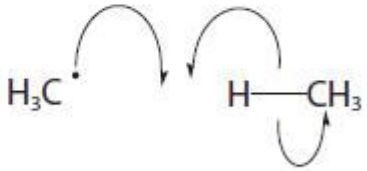
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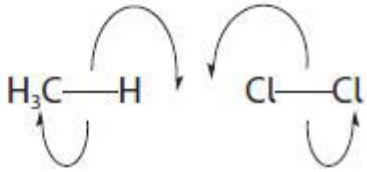
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☐ **A** 

☐ **B** 

☐ **C** 

☐ **D** 

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(Total for question = 1 mark)

Q21.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which pollutant **cannot** form when alkane fuels are burned in car engines?

☐ **A** hydrogen chloride
☐ **B** sulfur dioxide
☐ **C** carbon particulates
☐ **D** carbon monoxide

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(Total for question = 1 mark)

Q22.

The partial oxidation of butane in the presence of a catalyst has been investigated as a possible source of hydrogen.

$$C_4H_{10}(g) + 2O_2(g) \rightarrow 4CO(g) + 5H_2(g)$$

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(a) What is the most significant hazard associated with the pollutant CO(g)?

(1)

- ☐ **A** it is a major contributor to the formation of acid rain
- ☐ **B** it strongly absorbs infrared radiation
- ☐ **C** it is toxic at low concentrations
- ☐ **D** it is the main cause of ozone layer depletion

(b) A mixture of 40 cm^3 of butane and 20 cm^3 of oxygen reacts in a sealed system to form carbon monoxide and hydrogen.

What is the total volume of gas in the system at the end of the reaction, in cm^3?

(1)

- ☐ **A** 30
- ☐ **B** 60
- ☐ **C** 90
- ☐ **D** 120

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(Total for question = 2 marks)

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Q23.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is **not** a step in the reaction of chlorine with propane in ultraviolet radiation?

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- ☐ **A** $C_3H_7^{\bullet} + Cl^{\bullet} \rightarrow C_3H_7Cl$
- ☐ **B** $C_3H_7^{\bullet} + C_3H_7^{\bullet} \rightarrow C_6H_{14}$
- ☐ **C** $C_3H_7^{\bullet} + Cl_2 \rightarrow C_3H_7Cl + Cl^{\bullet}$
- ☐ **D** $C_3H_7^{\bullet} + HCl \rightarrow C_3H_7Cl + H^{\bullet}$

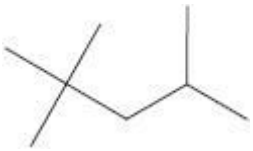
(Total for question = 1 mark)

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Q24.

The structure of a compound **X** is shown. **X** is present in petrol.



(a) (i) Give the molecular formula and IUPAC name for **X**.

(2)

.....

.....

(ii) Name the industrial process used to produce **X** from the straight-chain isomer.

(1)

.....

(b) Heptane is also present in petrol.

(i) Write an equation for the **complete** combustion of heptane. Include state symbols.

(2)

(ii) Identify, by name or formula, a pollutant that may result from the **incomplete** combustion of heptane.

(1)

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(c) Recently, sales of electric cars have increased.

Give **two** environmental benefits of using electric cars. Justify your answers.

(2)

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(Total for question = 8 marks)

Q25.

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Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

What is the number of structural isomers with the molecular formula C_5H_{12}?

- ☐ **A** six
- ☐ **B** five
- ☐ **C** four
- ☐ **D** three

(Total for question = 1 mark)

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Q26.

The reaction of methane with chlorine is a free radical substitution.

Which termination step does **not** occur?

- ☐ **A** $CH_3\bullet + CH_3\bullet \rightarrow C_2H_6$
- ☐ **B** $CH_3\bullet + Cl\bullet \rightarrow CH_3Cl$
- ☐ **C** $CH_3\bullet + H\bullet \rightarrow CH_4$
- ☐ **D** $Cl\bullet + Cl\bullet \rightarrow Cl_2$

(Total for question = 1 mark)

Q27.

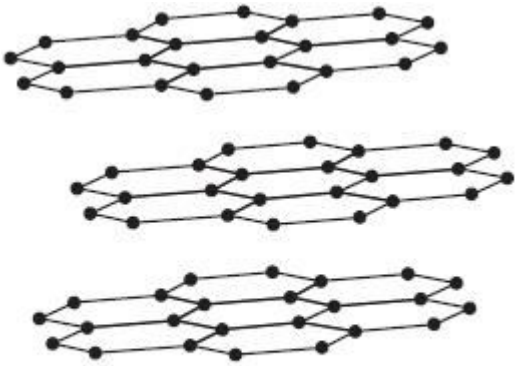
Graphite electrodes are used in the extraction of aluminium by the electrolysis of aluminium oxide, Al_2O_3, dissolved in a solvent (molten cryolite) at a temperature of 950 °C.

(a) Explain how the structure and bonding in graphite make it suitable for this application.

Refer to the diagram in your answer.

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graphite

(3)

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(b) The half-equations for the electrolysis reactions are shown.

Negative electrode $2Al^{3+} + 6e^- \rightarrow 2Al$

Positive electrode $3O^{2-} \rightarrow 1½O_2 + 6e^-$

The oxygen produced reacts with the graphite electrode.

$$C(s) + O_2(g) \rightarrow CO_2(g)$$

(i) Calculate the maximum volume, in dm^3, of CO_2, measured at r.t.p. which could be produced when 1.00 kg aluminium is extracted using this process.

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[The molar volume of a gas is 24.0 $dm^3\ mol^{-1}$ at r.t.p.]

(4)

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(ii) Every year in the United Kingdom, about 7.2 billion aluminium cans are recycled. Suggest three ways this recycling reduces energy consumption.

(3)

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(Total for question = 10 marks)

Q28.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Why does free radical substitution have limited use in industrial chemistry?

- ☐ **A** the reactions only occur in the upper atmosphere
- ☐ **B** initiation requires ultraviolet radiation
- ☐ **C** further substitution products are formed
- ☐ **D** termination reactions produce unwanted products

(Total for question = 1 mark)

Q29.

Oxygen is vital in the treatment of respiratory diseases. Oxygen is traditionally produced by the fractional distillation of air.

(a) Suggest **one** difference between the fractional distillation of air and of crude oil.

(1)

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(b) Hospital patients sometimes need to breathe air with a higher than normal concentration of oxygen.

The oxygen concentration can be increased to 90 % by passing dry air through a tube filled with zeolite which adsorbs most of the nitrogen.

(i) Dry air contains 21.0 % oxygen by volume.
The average human breath has a volume of 500 cm^3.
Calculate the volume of air, in dm^3, that would have to pass over the zeolite to obtain 500 cm^3 of gas containing 90 % oxygen by volume.

(2)

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(ii) Nitrogen molecules bind to zeolite using their outer electrons.
Draw a dot-and-cross diagram of the bonding in a nitrogen molecule.
Show outer electrons only.

(2)

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(c) Liquid oxygen is transported across the world.

Liquid oxygen has a density of 1140 g dm^{-3}.

(i) Calculate the volume of gas, in **m^3**, measured at room temperature and pressure (r.t.p.) that can be obtained from a bottle containing 13.9 dm^3 of liquid oxygen.

(4)

[Molar volume at r.t.p. = 24.0 dm^3 mol^{-1}]

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(ii) The total mass of the bottle is 80.0 kg when it is full of oxygen.
Calculate the mass of an empty bottle, using your working from (c)(i).
Give your answer to an appropriate number of significant figures.

(1)

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(iii) Explain why the container needs to have such a large mass.

(2)

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(Total for question = 12 marks)

Q30.

Many cars use fuels containing ethanol as well as hydrocarbons. The E numbers of these fuels show the percentage by volume of ethanol in the fuel. For example, an E5 fuel contains 5% ethanol and 95% hydrocarbons by volume.

(a) A brand of E10 fuel contains 92.2% hydrocarbons and 7.80% ethanol **by mass**.

The energy density is the amount of energy released per kg of a fuel burned.
The energy densities of the two components of this brand of fuel are shown.

energy density of hydrocarbons $= 46.5\ \text{MJ kg}^{-1}$

energy density of ethanol $= 29.7\ \text{MJ kg}^{-1}$

(i) Calculate the **mean** energy density of this E10 fuel in MJ kg^{-1}.
Give your answer to an appropriate number of significant figures.

(2)

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(ii) The density of the E10 fuel is 0.729 g cm^{-3}.
Calculate the mass of 1500 cm^3 of the fuel.

(1)

(iii) Calculate the energy released when 1500 cm^3 of the E10 fuel is burned in excess oxygen, using

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your answers from (a)(i) and (a)(ii).
[If you did not calculate a value for the **mean** energy density of the E10 fuel in (a)(i), use a value of 38.1 MJ kg^{-1}. This is not the correct value.]

(1)

(b) Many countries are gradually replacing E5 fuels with E10 fuels.

Explain how this change might result in a more sustainable fuel.

(3)

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(c) The straight-chain alkane hexane can be reformed to give a range of products. It was suggested that one such reaction may produce a cyclic hydrocarbon and hydrogen gas as the only products.
The cyclic hydrocarbon contains a ring of five carbon atoms.

(i) Write the equation for this reaction using molecular formulae. State symbols are not required.

(1)

(ii) Draw the structure of the cyclic hydrocarbon formed in this reaction.

(1)

(d) Hexane will react with chlorine in the presence of ultraviolet light to form a number of products.

(i) Describe the role of the ultraviolet light in the initiation step of this reaction.

(2)

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(ii) The alkane dodecane, $C_{12}H_{26}$, is formed in this reaction.
Write equations for the propagation step and for the termination step for the formation of this product.
Use molecular formulae in your equations.

(2)

Propagation step

Termination step

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(Total for question = 13 marks)

Q31.

The alkanes are a homologous series of saturated hydrocarbons.

(a) Draw the displayed formulae of the three alkanes with molecular formula C_5H_{12}.

(3)

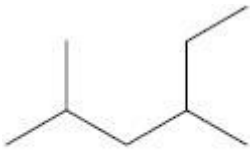
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(b) Give the systematic name of compound **P**.

(1)

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Compound **P**

Systematic name

.....

(c) The table shows the boiling temperatures of the first four straight-chain alkanes.

Molecular formula of alkane	Boiling temperature / °C
CH_4	−164
C_2H_6	−89
C_3H_8	−42
C_4H_{10}	−0.5

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Predict the molecular formula and boiling temperature of the straight-chain alkane that has five carbon atoms in its molecules.

(2)

Molecular formula .....

Boiling temperature .....

(d) Alkanes undergo incomplete combustion when they burn in a limited supply of air.

(i) Write the equation for the incomplete combustion of propane, C_3H_8, to form carbon, carbon monoxide, carbon dioxide and water.
State symbols are not required.

(1)

(ii) Explain the toxicity of carbon monoxide.

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(2)

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(e) Propane reacts with chlorine in the presence of ultraviolet radiation. The reaction starts when some chlorine molecules are split into free radicals. A mixture of products is formed.

(i) Write the two propagating steps to show how C_3H_7Cl is formed.
Curly arrows are not required.

(2)

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(ii) Identify the different C_3H_7Cl molecules that are produced in this reaction.

(1)

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.....

(iii) Give a reason why a mixture of C_3H_7Cl molecules is formed.

(1)

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(iv) Give a reason why some hexane is formed in this reaction.

(1)

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.....

(v) A small amount of a product with molar mass 113 g mol^{-1} is formed.
Deduce the structure and name of a possible product with this molar mass.

(2)

Structure

.....

Name

.....

(Total for question = 16 marks)

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Q32.

Which of these fuels is obtained from fermented sugar cane?

(1)

- ☒ **A** ethanol
- ☒ **B** hydrogen
- ☒ **C** petrol
- ☒ **D** propane

(Total for question = 1 mark)

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Q33.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

How many structural isomers have the formula C_6H_{14}?

- ☐ **A** 3
- ☐ **B** 4
- ☐ **C** 5
- ☐ **D** 6

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(Total for question = 1 mark)

Q34.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The substances formed from the combustion of petrol in car engines include

- ☐ **A** water, carbon dioxide and hydrogen
- ☐ **B** water, carbon monoxide and hydrogen chloride
- ☐ **C** water, carbon dioxide and sulfur dioxide
- ☐ **D** water, carbon particulates and hydrogen

(Total for question = 1 mark)

Q35.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which mixture could be formed when a **single** molecule of $C_{12}H_{26}$ is cracked?

- ☐ **A** butene, pentane and propene
- ☐ **B** hexane, butene and ethane
- ☐ **C** nonane and ethene
- ☐ **D** propene and decane

(Total for question = 1 mark)

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Q36.

Molecules of isooctane, found in petrol, have eight carbon atoms.

The skeletal formula of isooctane is shown.



(a) Give the molecular formula, empirical formula and IUPAC name for isooctane.

(3)

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Molecular formula.....

Empirical formula.....

IUPAC name.....

(b) Isooctane and octane are isomers that are both found in crude oil.
Octane has a boiling temperature of 125 °C, and isooctane has a boiling temperature of 99 °C.

This difference in boiling temperature can be used to separate the two isomers in a laboratory.
The technique is the same as that used to separate crude oil.

(i) Name the technique that could be used for this separation.

(1)

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(ii) Describe how this technique separates isooctane from octane.

(2)

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(c) Isooctane is added to petrol to increase its octane rating.

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Some high‑performance engines need fuel with a higher octane rating.

(i) Write the equation for the complete combustion of isooctane.
State symbols are not required.

(1)

(ii) Carbon monoxide and nitrogen monoxide are both pollutants produced in car engines.
Describe how each pollutant is formed in car engines, including the conditions required.
You may include appropriate equations.

(4)

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(iii) Identify another pollutant, other than carbon dioxide, carbon monoxide and nitrogen monoxide, produced in car engines.

(1)

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(d) Isooctane reacts with an excess of chlorine to form a mixture of chlorinated compounds containing the same number of carbon atoms as isooctane.

(i) State the type and mechanism of this reaction.

(1)

.....

(ii) Give the essential condition required.

(1)

.....

(iii) One of the chlorinated compounds contains 44.1% carbon and 6.9% hydrogen by mass. Calculate the molecular formula of this compound.

(3)

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(Total for question = 17 marks)

Q37.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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What is the best definition of a hydrocarbon?

- ☒ **A** a compound containing carbon, hydrogen and oxygen
- ☒ **B** a compound containing carbon and hydrogen only
- ☒ **C** a compound that contains single carbon to carbon bonds
- ☒ **D** a mixture containing carbon and hydrogen only

(Total for question = 1 mark)

Q38.

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Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Cycloalkanes undergo free radical substitution reactions by the same mechanism as methane. Three isomeric cycloalkanes are shown.



Isomer **X** Isomer **Y** Isomer **Z**

Which react with chlorine to produce four different monochlorinated products?

- ☒ **A** All 3 isomers
- ☒ **B** Isomer **X** and Isomer **Z** only
- ☒ **C** Isomer **Y** and Isomer **Z** only
- ☒ **D** Isomer **X** only

(Total for question = 1 mark)

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Q39.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

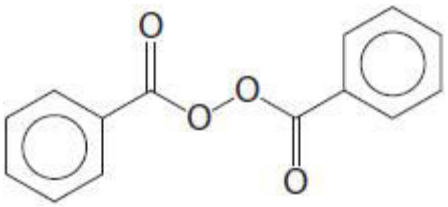
Polymerisation of alkenes occurs via a free radical mechanism.

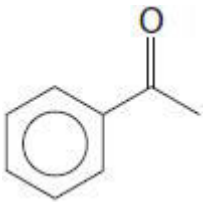
This reaction is started by the addition of small amounts of another compound.

The structure of one of these compounds is shown.

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This can be represented as R—O—O—R where R— is

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Which step in the mechanism is an initiation step?

☒ **A** $R—O—O—R \rightarrow 2R—O^{\bullet}$

☒ **B** $R—O—CH_2—CH_2^{\bullet} + CH_2=CH_2 \rightarrow R—O—CH_2—CH_2—CH_2—CH_2^{\bullet}$

☒ **C** $2R—O—CH_2—CH_2^{\bullet} \rightarrow R—O—CH_2—CH_2—CH_2—CH_2—O—R$

☒ **D** $R—O^{\bullet} + CH_2=CH_2 \rightarrow R—O—CH_2—CH_2^{\bullet}$

(Total for question = 1 mark)

Q40.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which statement about the cracking of crude oil is true?

The cracking of crude oil

☒ **A** separates the mixture into pure compounds

☒ **B** separates the mixture into fractions

☒ **C** separates saturated from unsaturated compounds

☒ **D** decreases the average number of carbon atoms per molecule

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(Total for question = 1 mark)

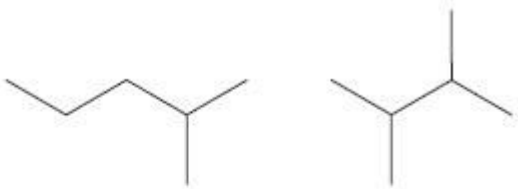
Q41.

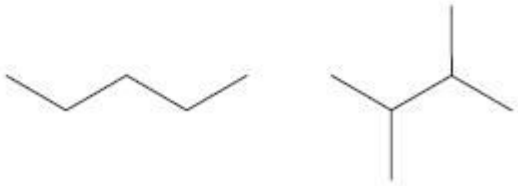
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

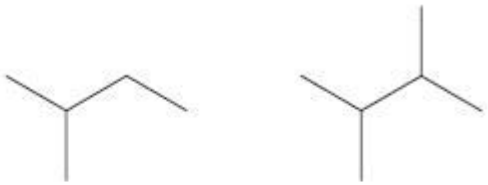
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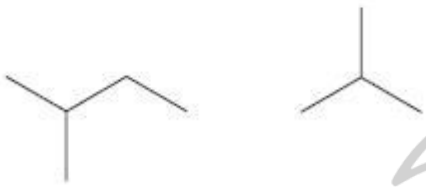
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Which structures represent a pair of isomers?

☒ A 

☒ B 

☒ C 

☒ D 

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(Total for question = 1 mark)

Q42.

This question is about alkanes and halogens.

Alkanes can react with halogens to form halogenoalkanes.

(a) Name the type and mechanism for the reaction between halogens and alkanes.

(1)

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(b) Chlorine reacts with butane.

(i) Give the equation for the initiation step.
Include appropriate arrows and electrons.

(2)

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(ii) Give an equation for each of the first two propagation steps.

(2)

First propagation step

.....

Second propagation step

.....

(iii) Give the equation for the termination step to form an alkane.

(1)

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.....

(c) Calculate the percentage yield if 1.00 g of trichlorobutane is produced from 10.0 g butane using the overall equation shown.

$$C_4H_{10} + 3Cl_2 \rightarrow C_4H_7Cl_3 + 3HCl$$

(3)

(d) Chlorine can be analysed in a mass spectrometer. On the grid shown, draw the spectrum you would expect to see for the molecular ion peaks for a chlorine molecule, Cl_2.

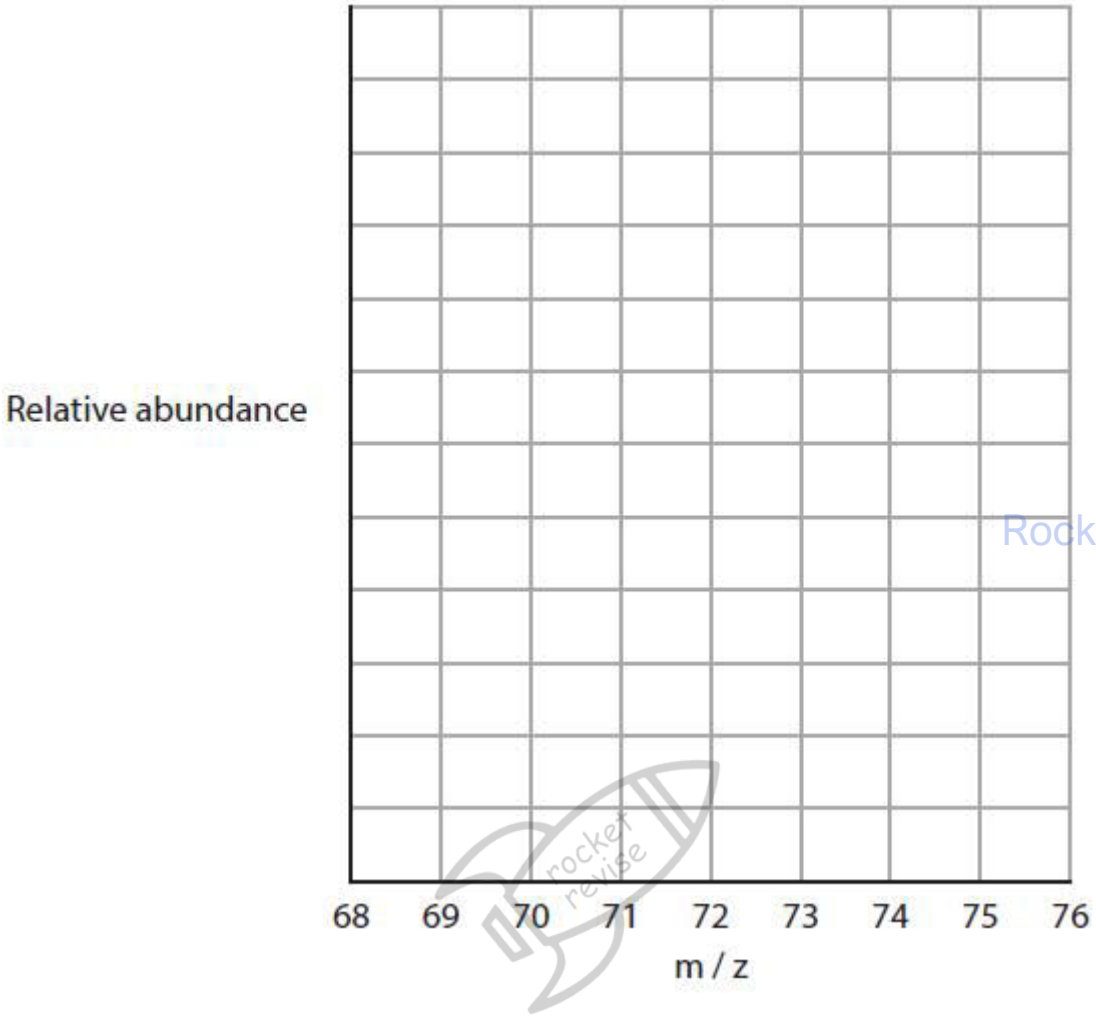
Isotopic abundances ^{35}Cl 75% $^{37}7Cl$ 25%

(2)

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(Total for question = 11 marks)

Q43.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which type of reaction occurs when an alkane is cracked?

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- ☐ **A** combustion
- ☐ **B** hydrolysis
- ☐ **C** neutralisation
- ☐ **D** thermal decomposition

(Total for question = 1 mark)

Q44.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

One molecule of a hydrocarbon is cracked to form two molecules of ethene, two molecules of propene and one molecule of propane. What is the molecular formula of the hydrocarbon?

- ☐ **A** C_8H_{18}
- ☐ **B** $C_{10}H_{22}$
- ☐ **C** $C_{11}H_{24}$
- ☐ **D** $C_{13}H_{28}$

(Total for question = 1 mark)

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Q45.

This question is about carbon dioxide.

(a) According to data from 2021, there are 415 ppm of carbon dioxide in the atmosphere by volume.

Calculate the moles of carbon dioxide present in 1.00 m^3 of air at 20.0 °C and 101 kPa.

[Ideal gas equation $pV = nRT$
$R = 8.31$ J mol^{-1} K^{-1}]

(4)

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(b) Dodecane $C_{12}H_{26}$, is found in kerosene and forms carbon dioxide during its complete combustion.

(i) State what is meant by complete combustion.

(1)

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(ii) Write a balanced equation for the complete combustion of dodecane, $C_{12}H_{26}$.
Include state symbols.

(2)

(iii) Kerosene is used as aeroplane fuel. A jet plane can carry a maximum of 800 passengers and uses 11 400 dm^3 of fuel per hour.
Calculate the mass, in kg, of carbon dioxide emitted from the engine per passenger on a full flight from Sydney to Hong Kong, flight time 9 hours 15 minutes.
Give your answer to three significant figures.
[Assume kerosene consists solely of $C_{12}H_{26}$
Density of dodecane = 0.749 g cm^{-3}]

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(6)

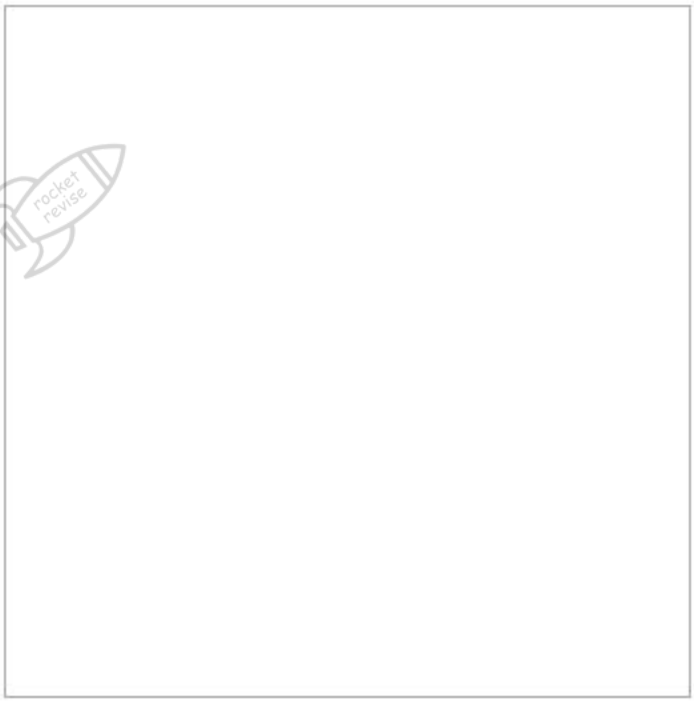
(iv) The formula $C_{12}H_{26}$ represents many isomers, including six diethyloctanes.
The names of four of these diethyloctanes are
3,3-diethyloctane, 3,4-diethyloctane, 3,5-diethyloctane, 3,6-diethyloctane.
Draw the **skeletal** formulae of the remaining two diethyloctanes.

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(2)

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(Total for question = 15 marks)

Q46.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

How many hydrogen atoms are in 14 g of methane?

[$L = 6.02 \times 10^{23}$]

- ☐ **A** 5.268×10^{23}
- ☐ **B** 6.880×10^{23}
- ☐ **C** 2.107×10^{24}
- ☐ **D** 2.752×10^{24}

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(Total for question = 1 mark)

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